

Universal Theorem Geometry I

Fibered Directed Multi-Geometry for Proof Search and MAPQUEST Navigation

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Abstract

This paper proposes the first layer of a universal geometry for formal theorem proving. The central claim is that theorem space should not be forced into a single metric model. Instead, a formal system contributes several mathematically distinct but interacting structures: zero-dimensional well-formed formulas; variable-dimensional inference charts whose dimension is rule arity; a directed proof-state geometry generated by legal inference; an address-dependent navigation geometry used by MAPQUEST; and translation maps between formal-system fibers. The resulting object is a tagged, fibered, directed multi-geometry.

The paper isolates two distances that must not be conflated. The first is an intrinsic directed proof distance: the infimum of legal transition costs from one proof state to another. The second is an extrinsic MAPQUEST address residual, including the rectilinear target distance previously defined from premise-slot addresses. The former is invariant under mere re-addressing, whereas the latter generally is not. This yields an address-distortion result and motivates a vector-valued geometric signature rather than a universal scalar distance. A family of navigation-efficiency quantities is then introduced, including the route efficiency

$$\rho_M(B, T) = \frac{d_F(B, T)}{L_M(B, T)},$$

which compares a prover’s successful route to the intrinsic proof geodesic. Finally, proof-preserving translations are shown to induce Lipschitz maps between theorem fibers when single-step simulation overhead is uniformly bounded. These constructions provide a rigorous basis for a series on Universal Theorem Geometry.

Revision 1.1 audit note. This revision narrows the stateization claim to monotone certificate-accumulating calculi, corrects the separation conditions for the directed proof distance, replaces an invalid recovery corollary by an explicit counterexample, makes the Lipschitz theorem precise for infimum distances, and requires inverse translations for the stated bi-Lipschitz comparison. No theorem from the previous version is retained solely by assertion when its proof did not justify its stated scope.

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1 Motivation

Automated theorem proving is often pictured as tree search, graph search, symbolic rewriting, or traversal of a syntactic state space. MAPQUEST adds a geometric navigation layer by assigning exact rational addresses to well-formed formulas when they occupy premise slots and by using rule-arity coordinate spaces to organize inference candidates. A separate graph-theoretic development identifies proofs with directed paths through proof states and makes shortest-path algorithms available as canonical proof-search benchmarks [1].

These two developments suggest a larger object, but they also expose a danger. Address proximity is not proof proximity. A formula can be close to another under a chosen numerical encoding while being deductively irrelevant; conversely, a deductively immediate inference can connect formulas whose numerical addresses are far apart. The rectilinear MAPQUEST target-distance framework explicitly warns that its positive values are properties of a frozen address embedding and do not, by themselves, bound proof length [2].

The goal of this paper is therefore not to declare one preferred metric on all theorems. The goal is to identify the structures that are genuinely present, state their invariance properties, and explain how they interact. The proposed universal object is a *fibred directed multi-geometry*: each formal system supplies a fiber, each fiber carries several layers of geometry, and suitable proof-preserving translations provide maps between fibers.

1.1 Guiding principle

The guiding principle is:

Universal theorem space is naturally a stratified, fibred, directed multi-geometry rather than a single metric space.

In particular, this paper keeps distinct:

1. the ontology of formulas;
2. the geometry of inference slots;
3. the intrinsic directed geometry of legal proofs;
4. the extrinsic geometry of search addresses;
5. the realized trajectory of a particular prover;
6. the distortion introduced by translations between formal systems.

2 Formal-system fibers

Definition 2.1 (Presented formal system). *A presented formal system is a tuple*

$$F = (\Sigma, \Gamma, R, V),$$

where Σ is a formal language, Γ is a designated hypothesis/axiom base, R is a set of finitary inference rules, and V is a verifier for finite certificates in the presentation. Let

$$W_F := \text{WFF}(F)$$

denote its set of well-formed formulas.

Convention 2.2 (Set-sized universe). *Fix a set-sized family $\mathfrak{F}$ of presented formal systems under discussion. This avoids proper-class coding ambiguities and is sufficient for every construction in this paper.*

Definition 2.3 (Universal tagged formula space). *The universal tagged formula space over $\mathfrak{F}$ is*

$$\mathcal{U}_0 := \bigsqcup_{F \in \mathfrak{F}} (\{F\} \times W_F).$$

The projection

$$\pi : \mathcal{U}_0 \rightarrow \mathfrak{F}, \quad \pi(F, \varphi) = F,$$

exhibits $\mathcal{U}_0$ as a family of fibers indexed by formal systems.

The tag F is essential. Two syntactically identical strings in different calculi are not silently identified. The symbol string alone does not determine which inference rules, axioms, proof costs, or verifier semantics are available.

Convention 2.4 (Zero-dimensional WFFs). *Each individual WFF is intrinsically a zero-dimensional location. Numerical addresses assigned to WFFs are navigation codes used when those WFFs occupy inference slots; an address does not turn a WFF into a positive-dimensional geometric object.*

This convention follows the repaired MAPQUEST framework [2] and is foundational for the dimensional bookkeeping below.

3 Variable-dimensional inference charts

Let $r \in R$ be a rule of arity k . A concrete use of r has an ordered premise tuple

$$(P_1, \dots, P_k)$$

and a conclusion C . The positive-dimensional geometry belongs to the premise slots of the inference application, not to the formulas in isolation.

Definition 3.1 (Exact address map and addressability scope). *For each $F \in \mathfrak{F}$ on which the MAPQUEST address geometry is invoked, assume W_F is at most countable and fix an injective exact address map*

$$\lambda_F : W_F \hookrightarrow \mathbb{Q} \cap [0, 1).$$

The intrinsic proof-state geometry developed below does not require this countability assumption; it is needed only for the rational address layer. The map may be generated from an injective syntactic enumeration $g_F : W_F \hookrightarrow \mathbb{N}$, for example

$$\lambda_F(A) = \frac{g_F(A)}{g_F(A) + 1}.$$

Definition 3.2 (Rule-arity chart). *For $r \in R_F$ with $\text{ar}(r) = k$, define its premise-slot chart by*

$$C_{F,r} := \{F\} \times \{r\} \times \mathbb{Q} \cap [0, 1)^k.$$

The inference-chart space of F is the tagged coproduct

$$\mathcal{C}_F := \bigsqcup_{r \in R_F} C_{F,r}.$$

The universal inference-chart space is

$$\mathcal{C} := \bigsqcup_{F \in \mathfrak{F}} \mathcal{C}_F.$$

A concrete inference candidate

$$c = (r; P_1, \dots, P_k; \xi)$$

has premise-address point

$$q_c = (\lambda_F(P_1), \dots, \lambda_F(P_k)) \in \mathbb{Q} \cap [0, 1]^k,$$

where ξ records finite witness, substitution, or instantiation data. Thus different rules of the same arity occupy distinct tagged components, while rules of different arities occupy different-dimensional components.

Remark 3.3 (Hilbert traversal). *A Hilbert or Hilbert-like traversal is best interpreted as a search-ordering mechanism inside a fixed finite-resolution approximation to $C_{F,r}$. It is a coordinate navigation policy, not the ontology of theoremhood and not the intrinsic proof metric introduced later.*

4 From inference hypergraphs to proof-state graphs

At formula level, a k -premise rule is naturally a directed hyperedge

$$(P_1, \dots, P_k) \longrightarrow C.$$

Shortest-path algorithms, however, operate most directly on ordinary directed graphs. The standard bridge is to move from formulas to proof states.

Definition 4.1 (Trusted proof state). *For the monotone, certificate-accumulating setting used in Part I, a proof state is a set $B \subseteq W_F$ of formulas accepted as available. A legal one-step transition*

$$B \longrightarrow B'$$

occurs when there is a verified rule application whose premises all lie in B and whose conclusion C is adjoined, with

$$B' = B \cup \{C\}$$

for a novelty-increasing step. Resource-sensitive, multiplicity-sensitive, or otherwise nonmonotone calculi require a richer state object (for example a multiset, sequent, or full proof configuration); the graph construction below then applies to those richer states rather than to plain sets of WFFs.

Definition 4.2 (State theorem graph). *Let $\mathcal{S}_F$ be the directed graph whose vertices are the reachable trusted proof states of F and whose labeled edges are legal verified transitions. The initial state is denoted B_0 , usually*

$$B_0 = \Gamma \cup \text{Ax}(F).$$

Proposition 4.3 (Stateization of monotone finitary inference). *For a finitary inference presentation in which premises persist and verified conclusions are accumulated monotonically, the directed inference hypergraph induces a directed proof-state graph in which each legal hyperedge application becomes an ordinary directed transition. Conversely, each labeled state transition records a concrete legal inference application.*

Proof. Suppose a rule application uses premises $P_1, \dots, P_k \in B$ to derive C , and the calculus retains the premises after the step. Then all required endpoints of the inference hyperedge are simultaneously available in B , so adjoining C produces the ordinary state transition $B \rightarrow B \cup \{C\}$. Conversely, by definition a legal labeled state transition is admitted only because a verified rule application was available at its source state; retaining the rule label and witness data recovers that concrete application. The proposition deliberately excludes calculi whose legal state evolution cannot be represented by monotone set extension. $\square$

This state construction is the mechanism by which multipremise proof structure becomes compatible with graph distance, breadth-first search, Dijkstra search, and A* search.

5 Intrinsic directed proof geometry

We now define the geometry that depends only on legal proof transitions and their chosen costs, not on WFF numerical addresses.

Definition 5.1 (Edge cost). *Assign each legal state transition e a cost*

$$w_F(e) \in [0, \infty).$$

The unit-cost model has $w_F(e) = 1$ for every edge. For a finite directed path

$$P = (e_1, \dots, e_n),$$

define

$$\text{cost}(P) := \sum_{j=1}^n w_F(e_j).$$

Definition 5.2 (Directed proof distance). *For proof states $B, C \in \mathcal{S}_F$, define*

$$d_F(B, C) := \inf\{\text{cost}(P) : P \text{ is a directed legal path } B \rightsquigarrow C\}.$$

If no such path exists, put $d_F(B, C) = \infty$.

Proposition 5.3 (Directed triangle law). *For all $A, B, C \in \mathcal{S}_F$,*

$$d_F(A, A) = 0$$

and

$$d_F(A, C) \leq d_F(A, B) + d_F(B, C).$$

In general $d_F(A, B)$ need not equal $d_F(B, A)$.

Proof. The empty path gives $d_F(A, A) = 0$. Concatenating an A -to- B path and a B -to- C path adds their costs, and taking infima yields the triangle inequality. Directionality of inference rules supplies no reason for symmetry. $\square$

Thus $(\mathcal{S}_F, d_F)$ is naturally an extended directed quasi-pseudometric (equivalently, a Lawvere-type directed metric structure): symmetry and point-separation are not automatic. For example, if every nontrivial edge cost satisfies a uniform lower bound $w_F(e) \geq \varepsilon > 0$, then any distinct state reachable by a nonempty path has forward distance at least ε . Without such a separation hypothesis, distinct states can in principle have distance 0 as an infimum even when every individual edge has positive cost.

Definition 5.4 (Target proof distance). *For a target WFF $T \in W_F$, let*

$$\mathcal{T}_F(T) := \{C \in \mathcal{S}_F : T \in C\}$$

be the target-state set. Define

$$d_F(B, T) := \inf_{C \in \mathcal{T}_F(T)} d_F(B, C).$$

In the unit-cost case this is the least number of legal state transitions needed to reach any state containing T .

Theorem 5.5 (Shortest-path recovery in the unit-cost case). *Assume the reachable portion of $\mathcal{S}_F$ is finite, or every reachable state has a finite effectively computable list of outgoing edges, and every edge has unit cost. If some target state containing T is reachable from B_0 , then breadth-first search from B_0 reaches a target at the least possible graph distance. With predecessor labels retained, tracing predecessors reconstructs a minimum-length state route and hence a verified proof certificate in the chosen state-transition model.*

Proof. This is the shortest-path ATP principle established in [1]. Breadth-first search explores vertices in nondecreasing graph distance. Finite branching implies that each finite depth layer is finite, so a reachable target at finite distance is eventually processed. The first processed target therefore has minimum graph distance. Predecessor labels recover the legal inference sequence along that state route. $\square$

Remark 5.6 (Weighted variants). *On finite graphs, or under the standard additional hypotheses guaranteeing termination on the reachable weighted graph, Dijkstra’s algorithm handles nonnegative edge costs. With an admissible heuristic and the usual A^* hypotheses, A^* may prioritize promising states while preserving optimality. These algorithms act on the intrinsic proof geometry and do not depend on a WFF address map.*

6 Extrinsic MAPQUEST address geometry

We now retain, separately, the address-based geometry used to organize inference candidates.

Definition 6.1 (Available premise configuration). *For a bank state $B \neq \emptyset$ and arity k , define*

$$\mathcal{B}_B^{(k)} := \lambda_F(B)^k.$$

For a concrete target-closing candidate

$$c = (r; P_1, \dots, P_k; \xi), \quad P_1, \dots, P_k \implies T,$$

let

$$q_c = (\lambda_F(P_1), \dots, \lambda_F(P_k)).$$

Definition 6.2 (Doorway rectilinear residual). *Define*

$$D_{\text{rect}}^{\lambda_F}(B, c) := \inf_{q \in \mathcal{B}_B^{(k)}} \|q_c - q\|_1.$$

Equivalently,

$$D_{\text{rect}}^{\lambda_F}(B, c) = \sum_{i=1}^k \inf_{A \in B} |\lambda_F(P_i) - \lambda_F(A)|.$$

Definition 6.3 (MAPQUEST rectilinear target distance). *For a nonempty represented family $\text{Pred}_F(T)$ of legal concrete target-closing inferences, define*

$$\text{MRTD}_{\lambda_F}(B, T) := \inf_{c \in \text{Pred}_F(T)} D_{\text{rect}}^{\lambda_F}(B, c).$$

This is the least rectilinear premise-slot displacement from currently available bank configurations to the represented legal doorways into T [2].

Proposition 6.4 (Monotonicity under bank growth). *Fix λ_F and the represented predecessor family $\text{Pred}_F(T)$. If $B \subseteq B'$, then*

$$\text{MRTD}_{\lambda_F}(B', T) \leq \text{MRTD}_{\lambda_F}(B, T).$$

Proof. For every arity k , $\lambda_F(B)^k \subseteq \lambda_F(B')^k$. Taking the infimum of distance to a fixed doorway over a larger comparison set cannot increase the value. Taking the outer infimum over target-closing candidates preserves the inequality. $\square$

6.1 Isolation and logical contact

Positive address distance is generally coding-dependent. The strongest logical contact occurs at isolated target addresses.

Definition 6.5 (Address isolation radius). *For $P \in W_F$, define*

$$\delta_{\lambda_F}(P) := \inf_{A \in W_F, A \neq P} |\lambda_F(P) - \lambda_F(A)|.$$

The address map is isolated at P when $\delta_{\lambda_F}(P) > 0$. For a k -premise candidate c , define

$$\delta_c := \min_{1 \leq i \leq k} \delta_{\lambda_F}(P_i).$$

Theorem 6.6 (Doorway eligibility threshold). *Assume $B \neq \emptyset$, $k \geq 1$, and λ_F is isolated at each required premise of c . If*

$$D_{\text{rect}}^{\lambda_F}(B, c) < \delta_c,$$

then every required premise of c is already in B . In particular,

$$D_{\text{rect}}^{\lambda_F}(B, c) = 0 \iff P_1, \dots, P_k \in B.$$

Proof. This is the isolation-radius argument of [2]. If some premise P_j is missing, every available $A \in B$ is at least $\delta_{\lambda_F}(P_j) \geq \delta_c$ away in the j th slot. Hence the total rectilinear residual is at least δ_c . $\square$

The theorem explains the precise interface between extrinsic navigation geometry and logic: positive residuals are generally heuristic, but crossing an isolation threshold becomes a certificate of premise eligibility.

7 Intrinsic and extrinsic geometry must be separated

The intrinsic proof distance is generated by legal inference. By contrast, the MAPQUEST target residual is generated by an address embedding. They answer different questions.

Quantity	Meaning
$d_F(B, T)$	Infimum of directed proof-state costs required to reach a state containing T ; a minimum when attained.
$\text{MRTD}_{\lambda_F}(B, T)$	Address-space displacement from the current bank to represented legal target-closing premise configurations.
$D_{\text{rect}}^{\lambda_F}(B, c)$	Address residual to one fixed legal target-closing doorway.

Proposition 7.1 (Re-addressing invariance of proof distance). *Let λ_F and μ_F be two injective exact address maps on the same presented formal system F , with the same proof-state graph and edge-cost function. Then*

$$d_F(B, T)$$

is identical under the two addressings.

Proof. The definition of d_F quantifies only over legal state transitions and their costs. Neither the set of legal paths nor their costs refers to λ_F or μ_F . $\square$

Proposition 7.2 (Address distortion). *Let $B \neq \emptyset$ be finite, let W_F be at most countable, and let a target-closing candidate c contain at least one premise $P_j \notin B$. Then there exist injective exact address maps λ_F and μ_F such that*

$$D_{\text{rect}}^{\lambda_F}(B, c) \neq D_{\text{rect}}^{\mu_F}(B, c),$$

while the intrinsic proof distance $d_F(B, T)$ is unchanged.

Proof. Let S be the finite set consisting of B together with the premises occurring in c , and choose $A_0 \in B$. Assign distinct rational addresses to the members of B inside a small interval near 0, and assign the same addresses under λ_F and μ_F to every member of S except P_j . Choose two unused rational values for P_j , one much closer to the finite set of bank addresses than the other. Then each slot occupied by P_j has a different contribution under the two addressings, while every slot whose premise is not P_j is unchanged. Because W_F is at most countable and only finitely many rational values have been prescribed, both partial assignments extend to injections of all of W_F into $\mathbb{Q} \cap [0, 1)$. Re-addressing does not alter the proof-state graph or its edge costs, so $d_F(B, T)$ is unchanged by Proposition 7.1. $\square$

If $\text{Pred}_F(T) = \{c\}$, then $\text{MRTD}_{\lambda_F}(B, T) = D_{\text{rect}}^{\lambda_F}(B, c)$, so the same construction explicitly changes MRTD itself while leaving intrinsic proof distance fixed.

Corollary 7.3 (Raw MRTD does not determine intrinsic proof distance). *There is no scalar function*

$$f : [0, \infty) \longrightarrow [0, \infty]$$

such that

$$d_F(B, T) = f(\text{MRTD}_{\lambda_F}(B, T))$$

for every presented system, bank state, target, represented predecessor family, and exact address map for which the two quantities are defined.

Proof. It suffices to give two proof problems with the same raw MRTD and different intrinsic proof distance. Consider a finite unit-cost system with atomic WFFs

$$A, P, Q, R, S, T, U$$

and rules

$$A \rightarrow P, \quad P \rightarrow T, \quad A \rightarrow Q, \quad Q \rightarrow R, \quad R \rightarrow S, \quad S \rightarrow U,$$

with no shorter rules to T or U . Let the initial bank be $B = \{A\}$, let the represented target-closing family for T contain only $P \rightarrow T$, and let that for U contain only $S \rightarrow U$. Then

$$d_F(B, T) = 2, \quad d_F(B, U) = 4.$$

Choose an injective address map with

$$\lambda_F(A) = \frac{1}{2}, \quad \lambda_F(P) = \frac{1}{4}, \quad \lambda_F(S) = \frac{3}{4},$$

and assign distinct unused rational addresses to the remaining WFFs. Because both target-closing rules are unary,

$$\text{MRTD}_{\lambda_F}(B, T) = \left| \frac{1}{4} - \frac{1}{2} \right| = \frac{1}{4} = \left| \frac{3}{4} - \frac{1}{2} \right| = \text{MRTD}_{\lambda_F}(B, U).$$

Thus one input value $1/4$ would have to satisfy both $f(1/4) = 2$ and $f(1/4) = 4$, which is impossible. $\square$

The corollary is stronger than re-addressing dependence alone: even with one fixed address map inside one formal system, equal raw address residuals can correspond to different intrinsic proof distances.

8 Geodesics, realized routes, and navigation efficiency

A verified route that attains the intrinsic distance is a geodesic of the directed proof geometry. A particular ATP need not return one, and with arbitrary weights the infimum need not be attained.

Definition 8.1 (Realized proof-route length). *Suppose a prover M settles T from state B and returns a verified state path $P_M(B, T)$. Define*

$$L_M^{\text{route}}(B, T) := \text{cost}(P_M(B, T)).$$

In the unit-cost model this is the number of transitions in the returned route.

Definition 8.2 (Route efficiency). *When*

$$0 < d_F(B, T) < \infty$$

and M returns a finite verified route, define

$$\boxed{\rho_M^{\text{route}}(B, T) := \frac{d_F(B, T)}{L_M^{\text{route}}(B, T)}}.$$

Proposition 8.3 (Bounds and optimality). *For every settled target in the domain of the definition,*

$$0 < \rho_M^{\text{route}}(B, T) \leq 1.$$

Moreover,

$$\rho_M^{\text{route}}(B, T) = 1$$

if and only if the returned route attains the infimum defining $d_F(B, T)$; equivalently, it is a minimum-cost route whenever that terminology is applicable.

Proof. By definition of d_F as an infimum over legal paths,

$$d_F(B, T) \leq L_M^{\text{route}}(B, T).$$

The domain assumption gives $d_F(B, T) > 0$, so the claimed range follows. Equality holds exactly when the realized route has cost $d_F(B, T)$ and therefore attains the infimum. $\square$

Definition 8.4 (Stretch and normalized route regret). *Define*

$$\sigma_M(B, T) := \frac{L_M^{\text{route}}(B, T)}{d_F(B, T)} = \frac{1}{\rho_M^{\text{route}}(B, T)}$$

and

$$R_M(B, T) := L_M^{\text{route}}(B, T) - d_F(B, T).$$

The normalized route regret is

$$\hat{R}_M(B, T) := \frac{L_M^{\text{route}}(B, T) - d_F(B, T)}{L_M^{\text{route}}(B, T)} = 1 - \rho_M^{\text{route}}(B, T).$$

The route metric measures proof quality, not search effort. A prover can eventually output a geodesic proof after exploring a vast number of irrelevant states.

Definition 8.5 (Search-expansion efficiency). *Let $E_M(B, T)$ be the number of state expansions performed before M first settles T . In the unit-cost setting, when $0 < d_F(B, T) < \infty$ and $E_M(B, T) > 0$, define*

$$\rho_M^{\text{search}}(B, T) := \frac{d_F(B, T)}{E_M(B, T)}.$$

Unlike ρ_M^{route} , this quantity compares an intrinsic path-length scale with an algorithmic exploration count. It is therefore a useful efficiency score but not a pure metric invariant. Its interpretation requires the state-expansion convention to be fixed across the systems being compared. No universal upper bound of 1 is asserted: depending on what counts as an expansion, $E_M(B, T)$ need not be comparable step-for-step with path length.

Definition 8.6 (Transaction efficiency). *If the common accounting unit is an SIC transaction, let $\tau_M(B, T)$ be the number of transactions consumed by M before settlement and let $d_F^{\text{txn}}(B, T)$ be the least possible transaction cost in the chosen transition model. Define*

$$\rho_M^{\text{txn}}(B, T) := \frac{d_F^{\text{txn}}(B, T)}{\tau_M(B, T)}.$$

These definitions separate three questions:

- $\text{MRTD}_{\lambda_F}(B, T)$: where should the navigation policy look?
- $d_F(B, T)$: how far away is the target deductively?
- $\rho_M(B, T)$: how efficiently did the prover navigate relative to the benchmark?

9 Cross-fiber maps and geometric distortion

A universal theory of theorem geometry must compare different formal systems without erasing their tags.

Definition 9.1 (Proof-preserving translation). *Let $F, G \in \mathfrak{F}$. A proof-preserving translation consists of a formula translation*

$$\tau : W_F \rightarrow W_G$$

together with a state map

$$\hat{\tau} : \mathcal{S}_F \rightarrow \mathcal{S}_G$$

such that $\hat{\tau}(B)$ contains the translations of the formulas in B and legal F -derivations are mapped to legal G -derivations of the translated conclusions.

Definition 9.2 (Uniform edge simulation bound). *The translation has a finite edge simulation constant $K \geq 0$ if every legal F -edge*

$$e : B \rightarrow B'$$

can be simulated in G by a directed path from $\hat{\tau}(B)$ to $\hat{\tau}(B')$ of cost at most

$$K w_F(e).$$

Theorem 9.3 (Lipschitz transport of proof distance). *If $\tau : F \rightarrow G$ is proof-preserving with finite edge simulation constant K , then for all states $B, C \in \mathcal{S}_F$ with $d_F(B, C) < \infty$,*

$$d_G(\hat{\tau}(B), \hat{\tau}(C)) \leq K d_F(B, C).$$

Proof. For every $\varepsilon > 0$, choose an F -path P from B to C with

$$\text{cost}(P) < d_F(B, C) + \varepsilon.$$

Replace each edge of P by its simulating G -path. Concatenation yields a G -path from $\hat{\tau}(B)$ to $\hat{\tau}(C)$ of cost at most

$$K \text{cost}(P) < K(d_F(B, C) + \varepsilon).$$

Taking the infimum over G -paths and then letting $\varepsilon \downarrow 0$ gives the claimed inequality. The case $K = 0$ is included. $\square$

Corollary 9.4 (Nonexpanding, bi-Lipschitz, and isometric translations). *If $K = 1$, $\hat{\tau}$ is nonexpanding on all pairs at finite F -distance. More generally, let $X \subseteq \mathcal{S}_F$ and $Y \subseteq \mathcal{S}_G$, and suppose*

$$\hat{\tau} : X \rightarrow Y$$

is a bijection whose inverse $\hat{\sigma} : Y \rightarrow X$ is induced by a proof-preserving reverse translation with finite constant $L > 0$. If $\hat{\tau}$ has constant K , then for all $B, C \in X$ with finite distances,

$$\frac{1}{L} d_F(B, C) \leq d_G(\hat{\tau}(B), \hat{\tau}(C)) \leq K d_F(B, C).$$

In particular, if $K = L = 1$, $\hat{\tau}$ is an isometry of the directed proof distances on X and Y .

Proof. The upper bound is Theorem 9.3. Applying the same theorem to $\hat{\sigma}$ gives

$$d_F(B, C) = d_F(\hat{\sigma}\hat{\tau}(B), \hat{\sigma}\hat{\tau}(C)) \leq L d_G(\hat{\tau}(B), \hat{\tau}(C)),$$

which yields the lower bound after division by L . $\square$

Remark 9.5 (Strict contractions and unit-cost rigidity). *If $0 \leq K < 1$, the translation is contractive on finite-distance pairs. However, suppose both fibers use separated unit-cost state graphs and $\hat{\tau}$ is injective on the endpoints of a source edge $B \rightarrow B'$. Then $d_F(B, B') = 1$, whereas the Lipschitz inequality would give*

$$d_G(\hat{\tau}(B), \hat{\tau}(B')) \leq K < 1.$$

A separated unit-cost target graph has only the values $0, 1, 2, \dots$ on reachable pairs, so the displayed distance must be 0, forcing $\hat{\tau}(B) = \hat{\tau}(B')$, a contradiction. Thus nontrivial strict contractions are naturally sought in weighted or normalized proof geometries, or in non-separated quotients, rather than between injectively embedded raw unit-step graphs.

This observation is important because it turns proof simulation overhead into a geometric distortion constant. Cross-system proof complexity can therefore be studied as geometry between fibers rather than by collapsing all systems into one undifferentiated space.

10 The universal multi-geometry

We can now package the construction.

Definition 10.1 (Universal Theorem Geometry, Part I). *Over a set-sized family $\mathfrak{F}$ of presented formal systems, define the Universal Theorem Geometry to consist of:*

1. *the tagged formula fibers $\{F\} \times W_F$;*
2. *the variable-dimensional inference-chart coproducts*

$$\mathcal{C}_F = \bigsqcup_{r \in R_F} \{F\} \times \{r\} \times \mathbb{Q} \cap [0, 1)^{\text{ar}(r)};$$

3. *the directed proof-state spaces $(\mathcal{S}_F, d_F)$;*
4. *for addressable fibers, one or more declared address embeddings λ_F and their induced MAPQUEST residuals;*
5. *realized prover trajectories and efficiency quantities such as ρ_M^{route} ;*
6. *a selected class of proof-preserving translations between fibers, equipped where possible with simulation-distortion bounds.*

Definition 10.2 (Universal geometric signature). *For a prover M , bank state B , and target T in F , define the partial signature*

$$\mathbf{D}_{F,M}^\lambda(B, T) = (d_F(B, T), \text{MRTD}_{\lambda_F}(B, T), m_T(B), L_M^{\text{route}}(B, T), \rho_M^{\text{route}}(B, T)),$$

whenever the listed quantities are defined. Here $m_T(B)$ is a declared discrete missing-premise residual over the represented predecessor family.

The point of **D** is not that all five coordinates are equally invariant. They are deliberately not. Rather, the signature records which aspect of theorem geometry is being measured.

Proposition 10.3 (No address-independent scalar can equal every raw address residual). *Assume a proof problem (F, B, T) admits two exact addressings λ_F and μ_F for which*

$$\text{MRTD}_{\lambda_F}(B, T) \neq \text{MRTD}_{\mu_F}(B, T).$$

Then no scalar quantity $D_F(B, T)$ that is independent of the address map can satisfy

$$D_F(B, T) = \text{MRTD}_{\nu_F}(B, T)$$

for every admissible exact addressing ν_F . In particular, an address-invariant intrinsic proof distance cannot simultaneously be identified with every raw MRTD value without additional canonicalization or compatibility hypotheses.

Proof. If such an address-independent scalar existed, the two addressings would give

$$D_F(B, T) = \text{MRTD}_{\lambda_F}(B, T) \quad \text{and} \quad D_F(B, T) = \text{MRTD}_{\mu_F}(B, T),$$

forcing the two distinct MRTD values to be equal, a contradiction. $\square$

11 A schematic model

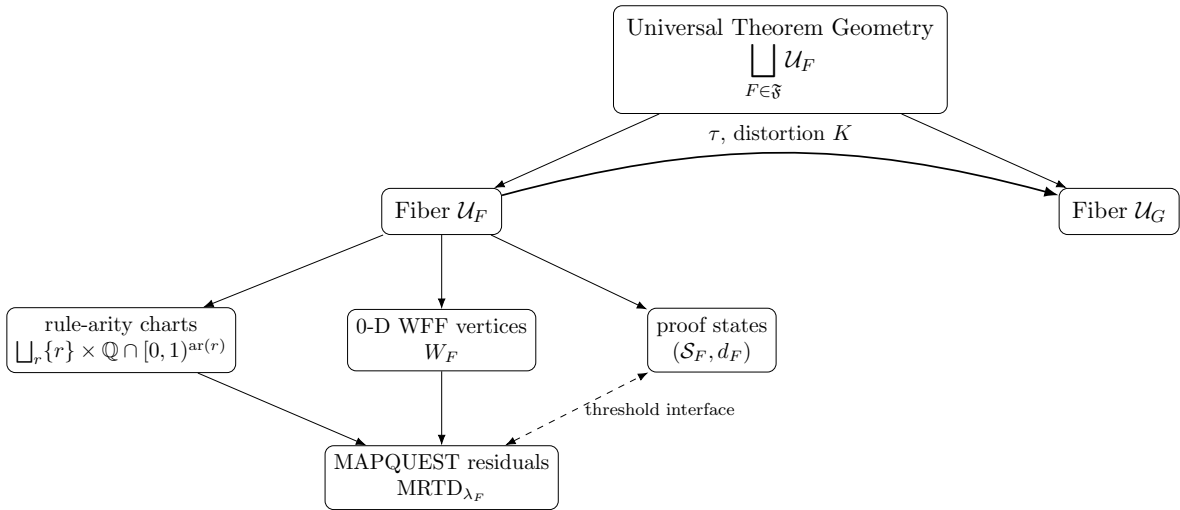


Figure 1: The Part I architecture. Positive-dimensional coordinates live in inference charts; intrinsic proof distance lives on proof states; MAPQUEST address residuals are retained as a separate navigation geometry. Proof-preserving translations act between formal-system fibers.

12 Research program opened by Part I

The present framework creates a sequence of concrete research problems rather than declaring the geometry finished.

Question 12.1 (Canonical addressing). *What additional structural constraints on λ_F make address residuals stable under reasonable recodings, or optimally correlated with intrinsic proof distance, without smuggling theoremhood into the address itself?*

Question 12.2 (Geodesic approximation). *For practical provers, how closely can MAPQUEST or AMLD search policies approximate proof geodesics under fixed resource bounds? The quantities ρ_M^{route} and ρ_M^{search} provide two distinct evaluation axes.*

Question 12.3 (Curvature and branching). *Can local branching, merge structure, bottlenecks, or proof-confluence phenomena be encoded by useful analogues of curvature or expansion in the directed state graph? Such notions should be derived from proof structure rather than imported by analogy alone.*

Question 12.4 (Quotient geometry). *Which equivalence relations on formulas, states, provers, or simulations produce well-defined quotient geometries? In particular, when does proof distance descend to a quotient without losing certificate recoverability?*

Question 12.5 (Cross-fiber equivalence). *Which proof-system simulations yield quasi-isometries, bi-Lipschitz equivalences, or genuine isometries between theorem fibers? This may provide a geometric language for comparing proof systems beyond theorem-set equivalence.*

Question 12.6 (Universal completion). *Is there a useful completion of a theorem fiber under a declared directed cost structure, and what would its boundary points represent? Any such completion must distinguish formal limit objects from actual derivable WFFs.*

13 Proposed series

This paper is intended as Part I of a series. A natural continuation is:

1. **Part I: Fibered Directed Multi-Geometry.** The present foundations: fibers, inference charts, directed proof distance, address residuals, efficiency, and translation distortion.
2. **Part II: Morphisms, Quotients, and Invariants.** Proof-preserving translations, simulations, quotient theorem spaces, bi-Lipschitz comparison, and invariants under recoding.
3. **Part III: MAPQUEST Navigation and Geodesic Approximation.** Hilbert-addressed search, admissible heuristics, regret, search efficiency, and empirical comparison with canonical shortest-path baselines.
4. **Part IV: Settlement Geometry for Multi-Agent Theorem Systems.** BANK growth, prover/refuter/independence/contradiction agents, verifier-gated settlement, and geometry of competing target classes.

14 Conclusion

A universal theorem space becomes mathematically cleaner when one stops asking for a single universal metric. The appropriate object is a tagged family of formal-system fibers. Inside each fiber, WFFs remain zero-dimensional; inference applications occupy rule-arity coordinate charts; proof states carry an intrinsic directed distance generated by legal inference; MAPQUEST contributes an extrinsic address-based navigation geometry; and actual provers trace measurable routes through this structure.

The two principal distances developed here are deliberately different. The intrinsic distance d_F measures deductive reachability. The MAPQUEST rectilinear target distance measures displacement in a chosen premise-slot embedding. Re-addressing leaves the first unchanged and can alter fixed-doorway residuals; for singleton or otherwise controlled predecessor families it can therefore alter MRTD as well. Thus no raw address metric can universally substitute for proof distance. Their interaction is nevertheless meaningful: under isolated addressing, sufficiently small doorway residual certifies that a target-closing inference is premise-eligible.

The efficiency quantity

$$\rho_M^{\text{route}}(B, T) = \frac{d_F(B, T)}{L_M^{\text{route}}(B, T)}$$

then evaluates a prover against the geometry rather than defining the geometry itself. Finally, proof-preserving translations with bounded step overhead act as Lipschitz maps between fibers, making simulation overhead a geometric distortion parameter.

The resulting picture is not a single Hilbert cube and not merely a theorem graph. It is a fibered, tagged, variable-dimensional directed hypergraph equipped with extended directed quasi-pseudometrics, address charts, navigation residuals, realized trajectories, and cross-fiber morphisms. That is the proposed starting point for Universal Theorem Geometry.

References

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